LRN · LAB REVENUE NAVIGATOR

BeechTree Three-Pillar Two-Agent Pipeline Cost Analysis

Technologies, where costs occur, and approximate monthly spend

IMPLEMENTATION COST BRIEF

Companion to the Foundry two-agent planning document
Workload · ~4–8 runs per month (weekly ES + backup)
Lean path · about $5–$20 / month
Standard path · about $190–$220 / month
16 August 2026 · USD list-price estimates

Internal use · Not a vendor quote · Verify on the Azure pricing calculator

Contents

This brief answers three questions: which technologies are required to implement the BeechTree Three-Pillar two-agent pipeline, whether they cost money, and approximately how much per month. Companion document: BeechTree_ThreePillar_Foundry_Agents_Plan.pdf. Section 8 is an index of cost terms.

1. Executive summary

Yes — there are Azure costs. For this design they stay small if you stay on pay-as-you-go. The only line that can jump is Logic Apps Standard hosting, which bills even when the pipeline is idle.

This pipeline runs about 4–8 times a month (BeechTree Executive Summary refresh plus a weekly backup). It is not a 24/7 chatbot. Microsoft does not charge extra to create a Foundry agent or write prompts. You pay for what it uses: model tokens, Logic Apps, storage, and secrets.

Path	When to use	Approx / month
Lean (recommended)	Weekly BeechTree only; Logic Apps Consumption + Functions Consumption	**$5 – $20**
Logic Apps Standard	Only if you need VNet or the Foundry Agent designer action	**$190 – $220**

Bottom line: technologies are Foundry + Logic Apps + Functions + Blob + Key Vault. Cost is real but small on weekly volume — unless you turn on Logic Apps Standard, which is where about $180/month appears even when the pipeline barely runs.

Figures are USD list-price ballparks for East US as of August 2026. They are not a Microsoft quote. Region, Enterprise Agreement discount, and model choice will move the number. Confirm on the Azure pricing calculator before creating the resource group.

2. Technologies required

Already in LRN — no new license

Piece	Role	New cost?
BeechTree_LRN SQL	Source of Three-Pillar numbers	**No**
Three-Pillar stored procedures	Agent 1 data (LIS, PMS, Cash)	**No**
ClaimLineCSVDataCapture STEP 14 / 16	“New Executive Summary” trigger	**No**
LabMetricsDashboard	Verify JSON vs the page	**No**
usp_NotesInsight_Insert	Store Agent 2 insights	**No**

Must add in Azure — billed

Technology	Role	Billed?
Azure subscription	Host everything	Account only
Microsoft Foundry (project + model)	Agent 2 insights	Yes — tokens
Azure Logic Apps	Schedule + chain Agent 1 → Agent 2	Yes
Azure Functions	Agent 1 JSON export	Yes, usually ~$0

Technology	Role	Billed?
Azure Blob Storage	JSON handoff file	Yes, pennies
Azure Key Vault	SQL connection string	Yes, pennies
Entra ID / Managed Identity	Authentication	Typically included
Application Insights	Run logs and failures	Small

Do not buy for this design

- Playwright or browser login — Agent 1 calls SQL, not the dashboard UI.
- Foundry hosted-agent containers — extra vCPU-hour; not needed if Agent 1 is a Function and Agent 2 is a native Foundry agent.
- Foundry visual workflows — retiring 1 December 2026.
- Azure OpenAI PTU (reserved capacity) — for high constant volume, not weekly runs.
- API Management — only later if you expose a public API.

3. Where the money is

Microsoft does not charge extra to create a Foundry agent or write prompts. You pay for consumption downstream of the weekly event.

Stage	Component	What is billed
Trigger	Logic Apps	Hosting (Standard) or per-action (Consumption)
Agent 1	Azure Function + SQL + Blob	Function executions (usually free grant) + tiny storage
Handoff	Blob Storage	A few KB–MB of JSON per week
Agent 2	Foundry model tokens	Input tokens (JSON + instructions) + output tokens (insights)
Optional tools	Bing, File Search, Code Interpreter	Skip unless you add them
Secrets / logs	Key Vault + Application Insights	Low fixed / usage

Watch-out: Logic Apps Standard bills for the server being on, not for weekly runs. Same 8 runs can cost pennies on Consumption or about $180 on Standard WS1. For this workload, prefer Consumption unless you have a hard requirement for Standard.

4. Lean path — recommended monthly estimate

Use Logic Apps Consumption + Functions Consumption + Foundry pay-as-you-go. The Logic App calls Foundry over HTTP (or a small Function), not a 24/7 Standard plan. Assumption: about 8 runs per month; JSON about 20–80 KB; Agent 2 on a GPT-4.1-class model (about $2 per 1M input tokens and $8 per 1M output tokens).

Where	What you pay for	Approx / month
Foundry Agent 2	Model tokens	**$1 – $8**
Logic Apps Consumption	Triggers + actions (first 4,000 actions free)	**$0 – $2**
Azure Functions	Agent 1 runs (1M executions free / month)	**~$0**
Blob Storage	JSON + insights files	**<$1**
Key Vault	Secrets + operations	**$1 – $3**
Application Insights	Logs	**$0 – $5**
Total lean	Weekly BeechTree pipeline	**$5 – $20**

Token example (Agent 2)

- One run at ~25,000 input tokens + ~3,000 output tokens ≈ $0.07.
- Eight runs ≈ $0.60 per month.
- A large JSON (~200,000 input tokens) is still only a few dollars a month at weekly volume.
- Start with a mid-size model. Switch to a mini model if quality is acceptable — that can cut token cost about 5–10×.

5. Logic Apps Standard path — the $180 line

Microsoft’s Foundry-in-the-designer path often wants Logic Apps Standard. That plan bills for reserved compute whether or not the workflow ran. WS1 in East US is typically about $180–$200 per month, plus the same small token and storage costs.

Where	Approx / month
Logic Apps Standard WS1 (always on)	**$180 – $200**
Foundry tokens + storage + Key Vault	**$5 – $15**
Total Standard	**$190 – $220**

Same 8 runs as the lean path. Most of the $200 is idle compute. Choose Standard only if you need VNet isolation, many workflows on one plan, or the native Foundry Agent designer action. Otherwise keep Consumption and invoke Agent 2 via HTTP.

6. What actually drives cost

Choice	Effect on bill
Weekly BeechTree only	Cheap. Tokens stay tiny.

Choice	Effect on bill
Logic Apps Standard	Largest bill. Avoid until you need VNet / many workflows.
Making Agent 1 a chat agent	Extra tokens and risk of invented numbers. Keep it a Function.
GPT-4o / GPT-5 vs mini	Mini can cut token cost about 5–10×. Start mid, drop to mini if quality is OK.
Hosted agents (containers)	Extra vCPU-hour. Not needed here.
Bing / File Search / Code Interpreter	Extra per search or per GB. Skip unless you add them.
More labs or daily runs later	Tokens scale with runs. Still usually under $50 until volume is high.

Existing SQL Server (BeechTree_LRN) and LabMetricsDashboard do not add Azure cost for this pipeline. Staff time to build Agent 1, the Logic App, and Agent 2 instructions is an internal labour cost, not an Azure meter.

7. Practical budget by phase

Phase	What to run	Expect / month
Pilot (1 lab, weekly)	Lean stack	**~$10**
Production BeechTree only	Same lean stack	**$10 – $25**
Same design, Logic Apps Standard	Only if you need that SKU	**~$200**
Later: more labs, daily runs	Tokens scale with runs	**Usually still under $50**

Kickoff decision

- Lock lean vs Standard before creating the resource group. That single choice is most of the monthly bill.
- Keep Agent 1 as an Azure Function (not a Foundry chat agent).
- Do not enable Bing, File Search, or hosted agents in the pilot.
- Set an Azure budget alert at $25 (lean) or $250 (Standard) so a misconfigured loop cannot surprise you.
- Confirm list prices in the Azure pricing calculator for your region and offer (PAYG vs Enterprise Agreement).

> *Recommended start: lean path. Create rg-lrn-threepillar-agents with Foundry (one model), Functions Consumption, Logic Apps Consumption, one Blob container, and Key Vault. Expect about $10/month while you prove Agent 1 JSON and Agent 2 insights.*

8. Index of cost terms

Alphabetical index of technologies and meters used in this brief.

Term	Cost context
Application Insights	Log store. Small monthly cost; often $0–$5 at this volume.
Azure Blob Storage	Handoff for three-pillar.json. Pennies per month.
Azure Functions Consumption	Agent 1 runtime. 1 million executions free per month; this pipeline is far below that.
Azure Key Vault	Holds BeechTree SQL connection string. About $1–$3/month.
Azure subscription	Required account. No extra fee just to exist if resources are pay-as-you-go.
Entra ID / Managed Identity	Auth from Function and Logic App. Typically no extra meter.
Foundry Agent Service	No extra charge to create/run native agents; you pay model tokens and optional tools.
Foundry hosted agents	Container vCPU/memory hours. Do not use for this design.
Foundry model tokens	Main AI cost. Input + output billed separately (example GPT-4.1 ~$2 / $8 per 1M).
Logic Apps Consumption	Pay per action. First 4,000 actions free. Right SKU for weekly runs.
Logic Apps Standard WS1	Always-on plan ~$180–$200/month. Only if VNet or designer Agent action is required.
PTU (provisioned throughput)	Reserved model capacity. Not needed at weekly volume.
SQL BeechTree_LRN	Existing on-prem / lab database. No incremental Azure cost.
Token (input)	JSON + system instructions sent to Agent 2.
Token (output)	Insights JSON Agent 2 writes back.

End of cost brief. Next action: choose lean vs Standard, then begin Phase 0 in the planning PDF.